

Major Features 0.3.0

Multi-period NameLists

The code now uses separate DecadeData and NameList structures to handle names for up to seven different periods (decades), reflecting historical data segmentation.

Interactive Family & Couple Generation

The CLI and interactive modes support generating not just single names, but couples and families (including parents + children), with randomization of number of children and birth years.

Flexible Configuration Loading

The config file can override the default data paths for each category of names via load_config.

If the config file is missing, sensible defaults are used.

Verbose mode prints information about what was loaded.

Period-based Name Loading

New function load_names_period_list loads names from CSV files, treating each column as a different period, enabling the generation of historically plausible names.

Input Validation and Cleanup

Better trimming, input validation, and memory management of loaded names.

1. Options and Parameters

New/expanded options handled:

- p (period selection), now with a range 1–7, mapped to different decades.
- g (gender), now ignored in couple/family modes.
- P (couple) and -F (family).
- A (age/birth year simulation).
- k/--kids and random kids.
- S (set specific surname override).
- L (exclude surname).
- m, -M for middle name customization.

Many of these parameters are also prompted in interactive mode, with sensible defaults and input validation.

2. Verbose Output

When verbose, after file loading, the code prints the count of names per period for the loaded CSVs (e.g., male/female first and middle names for period 7).

3. Internationalization and Windows Support

Sets the console encoding for Unicode on Windows.

Uses setlocale(LC_ALL, "").

4. Help Message Updates

The CLI help text now reflects all new options, including --family, --age, --middle-chance, --max-middle-names, etc.

5. Memory Management

Proper freeing of dynamic memory for name lists, config paths, and cleaning up at exit.

Summary

Main new features:

- Period/decade-based name generation: More historically accurate names.

- Family and couple generator: Can simulate parents/children with customizable parameters.

- Configurable and more robust file loading.

- Interactive mode with much richer option prompts.

Improvements:

- Verbose stats.

- Better input validation and memory cleanup.

- Help/version updated for new functionality.

Would you like a more detailed comparison with a specific previous version, or an explanation of a particular feature in this file?